MIL-C-14892(MU)
1 November 1973
SUPERSEDING
FA-PD-MI-2524, Rev. I
24 April 1964

MILITARY SPECIFICATION
CASE, CARTRIDGE, 105MM, M150B1

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Case, Cartridge, 105MM, M150B1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Armament Command

F10521648	- Case, Cartridge, 105MM, M150B1
F7548235	- Cartons, Ammunition for Interplant Shipment of Steel Cartridge Cases, 37MM to 120MM, Inclusive
IEL-10521648	- Inspection Equipment List for Case, Cartridge, 105MM, M150B1

FSC 1315

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

ASTM Method - American Standards for Testing and Materials

(Application for copies should be addressed to the American Standars for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-The cartridge case shall comply with all requirements of the applicable drawings, referenced specifications and the following:

3.2 Material.-The supplier shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Thermal treatment.-The entire cartridge case shall be subjected to a thermal treatment after the heading operation, by heating above the upper critical temperature point, quenching and tempering to the desired properties.

3.4 Stress relief.-The cartridge case shall be adequately stress relieved at the times and temperatures suitable for development of desired properties.

3.5 Mechanical properties.-The mechanical properties specified on the applicable drawing and referenced specifications shall be uniform throughout each section.

3.6 Protective coating.-Protective coating as specified on the applicable drawing and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration.

3.7 Maximum profile.-Maximum profile of the cartridge case shall comply with the requirements of the applicable drawing.

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3.8 Functional.-The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (For Information Only)

3.8.1 Obturation.-The cartridge case shall effectively seal the breech of the weapon against leakage of gases generated during firing.

3.8.2 Soundness.-After firing, the cartridge case shall be free of cracks or splits, except as permitted by Table III.

3.8.3 Extraction.-After firing, the cartridge case shall be extracted by the service exactors when the weapon is operated in the normal manner (see 6.3).

3.8.4 Primer retention.-The cartridge case shall retain its primer during firing and extraction.

3.9 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects.-The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches or porosity.

3.9.2 Flow of metal.-The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.9.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.9.4 Burr.-No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the item.

3.9.5 Foreign matter.-No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter during inspection or testing and prior to shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 274 cartridge cases as follows:

254 Cartridge Cases - Coated
50 Cartridge Cases - Uncoated

250 of the above coated cartridge cases shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, material used, drawing specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and and AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as cracks, burr and foreign matter. The roughness comparison specimens prescribed in ASA B46.1 shall be used as basis for surface roughness determination.

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4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during first article inspection and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Contractor will immediately institute corrective action whenever a defective tool control dimension is found.

4.3.2.2 Case, Cartridge, Dwg. F10521648 - prior to coating.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	True position of bead with mouth	Gage
102.	Diameter of flange, min.	Gage
103.	Diameter of mouth	Gage
104.	Diameter of bead	Gage
105.	Length of mouth diameter, min.	Gage
106.	Thickness of flange, min.	Gage
107.	Thickness of head, min.	Gage
108.	Thickness of wall at basic location near head	Gage
109.	Thickness of wall at basic location near mid body	Gage
110.	Thickness of wall at basic location near mouth	Gage
111.	Thickness of wall at mouth	Gage
112.	Contour of inside of head and sidewall, not smooth	Visual
113.	Metal defective (except as otherwise classified)	Visual
MINOR		
201.	Diameter of body at flange, min.	Gage
202.	Location of first basic diameter from head end, min.	Gage
203.	Location of second basic diameter from head end, min.	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
204. Location of third basic diameter from head end, min.	Gage
205. Location of fourth basic diameter from head end, min.	Gage
206. Length total, min.	Gage
207. Location of bead	Gage
208. Radii or chamfers, missing or incorrect	Visual (see 6.2)
209. Surface finish, improper	Visual (see 6.2)
210. Burr	Visual
211. Foreign matter	Visual

4.3.2.3 Case, Cartridge, Dwg. F10521648 - after coating.

CRITICAL

1. Crack(s) in sidewall within 8 inches of head	Visual - Four Power (4X) Magnification

MAJOR

101. True position of pitch diameter of thread with body	Gage
102. True position of primer hole counterbore with pitch diameter of thread	Gage
103. Pitch diameter of thread, max.	Gage
104. Minor diameter of thread, max.	Gage
105. Length total, max.	Gage
106. Depth of primer hole counterbore	Gage
107. Protective coating inadequate or damaged	Visual

MINOR

201. Pitch diameter of thread, min.	Gage
202. Diameter of primer hole counterbore	Gage
203. Dent or damaged	Visual
204. Thread damaged or not full	Visual
205. Silicone compound missing from thread or primer hole counterbore	Visual
206. Marking missing, incorrect or illegible	Visual

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4.3.2.4 Carton, Dwg. F7548235 - prior to sealing.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Incorrect number cartridge cases in carton	Visual
203. Separator missing or damaged	Visual

4.3.2.5 Carton, Dwg. F7548235 - after sealing.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton improperly sealed	Visual
203. Contents loose	Visual/Manual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Maximum profile gage (a major defect).-Each cartridge case in the lot shall be gaged in a Government certified maximum profile gage. Failure of the cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the case.

4.3.4 Inspecting testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.4.1 Thermal treatment (a major defect).-Each cartridge case shall be subjected to a hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.3 or indications that the cartridge case failed to harden shall be cause for rejection of the cartridge case.

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4.3.4.2 Mechanical properties.

4.3.4.2.1 Hardness.-Twenty cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with 3.5 shall be cause for rejection of the sample lot.

4.3.4.2.2 Tension test.-Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test in accordance with the requirements of the applicable drawing. The results shall be recorded for information only. The cartridge cases used for the hardness test may be used for this test.

4.3.4.3 High stress expansion test.-Each cartridge case in the lot after heat treatment but prior to tapering shall be subjected to this test. Evidence of cracks in any portion of the case prior to or after tapering shall be cause for rejection of the case.

4.3.4.4 Protective coating.-The sample size for the examination and testing the protective coating except thickness of coating and salt spray test shall be as specified in the applicable specification. Failure of any specimen to comply with the applicable requirements shall be cause for rejection of the lot represented by the failing specimen.

4.3.4.4.1 Thickness of coating.-Two cartridge cases from the initial production sample and two cartridge cases from each 5000 cases or fraction thereof produced shall be subjected to this test. Failure of any specimen to comply with the requirements of the applicable specification shall be cause for rejection of the lot represented by the failing specimen.

4.3.4.4.2 Salt spray.-Two cartridge cases from the initial production sample and two cartridge cases from each 5000 cases or fraction thereof produced shall be subjected to this test until 5 consecutive days production have successfully passed the test, then sampling can be reduced to one cartridge case per each 5000 cases or fraction thereof. In no case shall less than 4 cartridge cases per shipping lot be tested. Failure of any specimen to comply with the requirements of the applicable drawing shall be cause for rejection of the lot represented by the failing specimen. If when on reduced sampling a failure occurs, corrective action shall be taken and sampling shall revert to the original sample size until 5 consecutive days production have again successfully passed the test.

4.3.4.5 Macro examination.-Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be examined for folds and cold shuts. Failure of one or more cartridge cases to comply with 3.9.1.2 shall be cause for rejection of the sample or lot. The cartridge cases used for the hardness test may be used for this test.

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4.3.4.6 Functional.-The following proving ground tests will be performed by the Government:

4.3.4.6.1 Initial production sample.-The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table II.

TABLE I

Test Phase	Quantity to be Tested
Excess Pressure Phase	100
Cold Temperature Phase	50
Excess, Excess Pressure Phase (For Information Only)	100

TABLE II

Defect Characteristic	Sample Accept
1. Obturation	0
2. Splits or cracks within 1 caliber of the head	0
3. Splits or cracks in sidewall more than 1 caliber from the head	1
4. Extraction	0
5. Primer retention	0

4.3.4.6.2 Production lot sample.-One hundred cartridge cases from each production lot of 25,000 or less, will be ballistically tested in the excess pressure phase. (When lots exceed 25,000 cartridge cases, 4 additional cartridge cases will be tested for every 1000 or fraction thereof, above 25,000.) Acceptance and rejection criteria shall be in accordance with Table III. The lot shall be considered eligible for retest, using double the number of samples, provided that after firing, the rejection criteria of Table III has not been attained.

TABLE III

Defect Characteristic	First Sample Accept	First Sample Reject	Cumulative Sample Accept	Cumulative Sample Reject
1. Obturation	0	2	1	2
2. Splits or cracks within 1 caliber of the head	0	1	Not Permitted	
3. Splits or cracks in sidewall more than 1 caliber from head	1	3	2	3
4. Extraction	0	2	1	2
5. Primer retention	0	2	1	2

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4.3.7 Inspection equipment.-Index of Inspection equipment list IEL 10521648 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Thermal treatment.-This test shall be performed on a hardness comparator of a nondestructive type approved by the contracting officer.

4.4.2 Mechanical properties.

4.4.2.1 Hardness.-This test shall be performed on the locations specified by the applicable drawing in accordance with the procedures prescribed for Rockwell Hardness Testing by ASTM Method E-10.

4.4.2.2 Tension test.-The test specimens shall be taken from the cartridge case in accordance with the standard position and size of tensile specimen chart, Figure I, the yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed for these tests by ASTM Method E-8.

4.4.3 High stress expansion test.-The cartridge case, after heat treatment shall be expanded diametrically 0.05 inches minimum while under pressure in a zone extending from 1 ½ inches above the flange to 9 inches above the flange, expansion from 1 ½ inches above the flange will decrease as the flange is approached, expansion at flange to be zero.

4.4.4 Protective coating.-The criteria specified by the applicable specification shall be utilized in determining acceptability of the protective coating.

4.4.4.1 Thickness of coating.-The thickness of the protective coating may be determined by either method specified in the applicable specification.

4.4.4.2 Salt spray.-The samples provided by 4.3.4.4.2 shall be tested using the criteria and equipment specified for this test by ASTM Method B-117. The significant inner surface as well as the significant outer surface of the case will be examined for compliance with the requirements of the applicable specification.

4.4.5 Maximum profile gage.-The cartridge case, after application of protective coating shall freely enter the Government certified maximum profile gage. The gate of the gage shall close without interference with the cartridge case head.

4.4.6 Macro examination.-The cartridge case shall be cut in a 90 degree section along the axis through the primer hole. The specimen shall be etched in accordance with the procedures prescribed for this test by ASTM Method 317.

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4.4.7 Functional.

4.4.7.1 Initial production sample.

4.4.7.1.1 Excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit (F) to plus 75 degrees F, and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.1.2 Cold temperature phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 24,200 pounds per square inch (PSI) to 28,600 PSI piezo equivalent pressure, when temperature conditioned at 70 degrees F. The complete round will be temperature conditioned at minus 35 degrees F, to minus 45 degrees F, and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.1.3 Excess, Excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F, to plus 75 degrees F, and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.2 Production lot.

4.4.7.2.1 Excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F, and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.-Not applicable.

5.1.2 Level B.-Not applicable.

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5.1.3 Level C.-Cartridge cases shall be packed in accordance with the requirements of Drawing F7548235.

5.2 Packaging and packing.-Packaging, packing and marking shall be in accordance with MIL-STD-1169.

5.3 Data card.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Prior to fabrication of an initial production sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to five process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.2 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Extraction.-The term "extraction" shall mean that the fired cartridge is loosedned and pulled to the rear of the weapon sufficeintly to permit removal by hand.

6.4 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.5 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2524, Rev. 1 dated 24 April 1964.

Custodian:

Army-MU

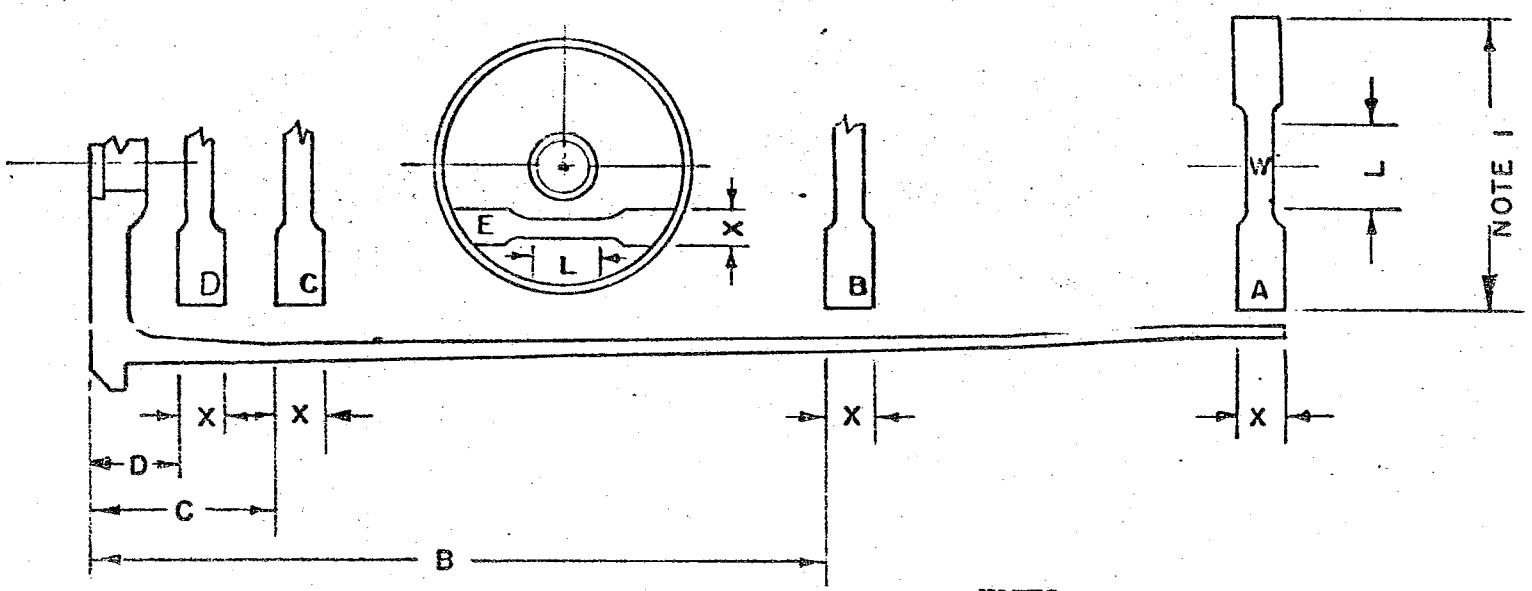
Preparing activity

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Project No.1315-A248

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STANDARD POSITIONS AND SIZES OF TENSILE SPECIMENS FOR CASE, CARTRIDGE, 105MM, M150B1



SPECIMEN	X	W	L	LOCATION FROM HEAD
A	.40	.25	1.0	AT MOUTH
B	.40	25	1.0	15 25
C	.40	.25	1.0	2.00
D	.40	.25	1.0	0.75
E	.40	.25	1.0	IN HEAD

FIGURE I

NOTES:

1. Overall length of specimen to suit fixture in testing machine.
2. Specimen C and D to be machined to uniform thickness parallel to inside surface.
3. Centerline of specimen E to be 1.75 inches from centerline cartridge case.
4. Specimen E to be machined so as to eliminate effects of stamping on test.
5. Yield strength in 0.2 percent offset.
6. Percent elongation in 1.0 inch gage length.

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